

Figure 3: ARGoS basic simulation

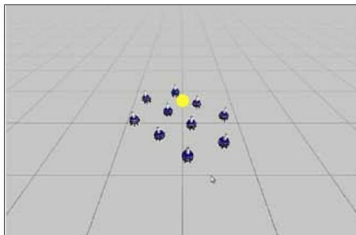


Figure 4: Flock formation in ARGoS

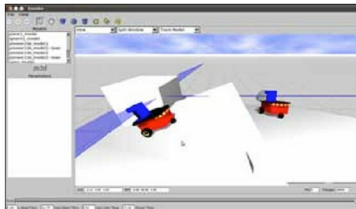


Figure 5: Two pioneer robots in Gazebo

The ARGoS simulator

ARGoS (Autonomous Robots Go Swarming) is a multi-robot simulator designed to support large teams of robots. Its design is pretty different from the design of other simulators. Its most distinctive feature is that the 3D simulated world can be divided in regions, and each region can be assigned to a different physics engine. Furthermore, ARGoS' design revolves around the concept of tunable accuracy. In other words, in ARGoS, everything is a plug-in (robot models, sensors, actuators, physics engines, visualisations, etc) and the user can select which plug-ins to use for an experiment. Since different plug-ins have different accuracy and

computational costs, users can choose which plug-ins to use for each aspect of the simulation and assign resources only where it matters. This makes the simulation as fast as possible.

At the time of writing, ARGoS supports the Swarmanoid robots (foot-bot and eye-bot) and the e-puck. ARGoS supports Linux and Mac OSX. Binary packages are available for Ubuntu, Slackware and Mac OS-X. In addition, a generic binary package can be used for other Linux distributions. Everything is downloadable from iridia.ulb.ac.be/argos/download.php.

Robot control code for ARGoS is written in C++. Experiments are configured through an XML file. To run a demo simulation in ARGoS, download the examples from the same URL, unzip the archive, and run the experiment with the following command:

```
launch_argos -c xml/diffusion_1.xml
```

That results in a screen popping up, as shown in Figure 3, with a single foot-bot robot in a red box-like environment.

Compared to Stage, ARGoS provides a 3D simulation environment. In addition, since the physics engines can be chosen by the user, any kind of experiment is possible, including complex self-assembly. Its performance is found to be superior to Stage's. With the full power of four cores on a normal desktop PC, ARGoS can simulate more than 4000 robots in real-time.

Watch out for Part 3 of this series, which will contain a comprehensive discussion on ROS. **END** 

Reference

- [1] I Robot, 2004 movie, www.imdb.com/title/tt0343818/
- [2] R Vaughan, 'Massively Multiple Robot Simulations in Stage', *Swarm Intelligence* 2(2-4):189-208, 2008. Springer.
- [3] N Koenig and A Howard, 'Design and Use Paradigms for Gazebo, An Open-Source Multi-Robot Simulator', a paper presented at the IEEE/RSJ International Conference on Intelligent Robots and Systems, 2004
- [4] Carlo Pinciroli, Vito Trianni, Rehan O'Grady, Giovanni Pini, Arne Brutschy, Manuele Brambilla, Nithin Mathews, Eliseo Ferrante, Gianni Di Caro, Frederick Ducatelle, Timothy Stirling, Álvaro Gutiérrez, Luca Maria Gambardella and Marco Dorigo, 'ARGoS: a Modular, Multi-Engine Simulator for Heterogeneous Swarm Robotics.' – proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2011), pages 5027-5034, IEEE Computer Society Press, Los Alamitos, CA.

By: A Bhaumik

The author is a faculty member at the Birla Institute of Technology, Mesra, Ranchi. He is a robotics enthusiast and his ramblings on mobile robots can be found at his blog, robotica.blogspot.in/. He can be reached at arkapravobhaumik@gmail.com.